Hyperspectral Image Classification Based on Deep Attention Graph Convolutional Network

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***Abstract*— Hyperspectral images (HSIs) have gained high spectral resolution due to recent advances in spectral imaging technologies. This incurs problems, such as an increased data scale and an increased number of bands for HSIs, which results in a complex correlation between different bands. In the applications of remote sensing and earth observation, ground objects represented by each HSI pixel are composed of physical and chemical non-Euclidean structures, and HSI classification (HIC) is becoming a more challenging task. To solve the above problems, we propose a framework based on a deep attention graph convolutional network (DAGCN). Specifically, we first integrate an attention mechanism into the spectral similarity measurement to aggregate similar spectra. Therefore, we propose a new similarity measurement method, i.e., the mixed measurement of a kernel spectral angle mapper and spectral information divergence (KSAM-SID), to aggregate similar spectra. Considering the non-Euclidean structural characteristics of HSIs, we design deep graph convolutional networks (DeepGCNs) as a feature extraction method to extract deep abstract features and explore the internal relationship between HSI data. Finally, we dynamically update the attention graph adjacency matrix to adapt to the changes in each feature graph. Experiments on three standard HSI data sets, namely, the Indian Pines, Pavia University, and Salinas data sets, demonstrate that the DAGCN outperforms the baselines in terms of various evaluation criteria. For example, on the Indian Pines data set, the overall accuracy of the proposed method achieves 98.61% when the training sample is 10%.**

***Index Terms*— Attention mechanism, graph convolution network (GCN), hyperspectral image classification (HIC), similarity measurement.**

I. Introduction

HYPERSPECTRAL images (HSIs) have drawn much attention in recent decades due to their rich data and high spectral resolution [1], [2]. By leveraging such rich hyperspectral information, HSIs have been successfully applied to a wide range of applications, including urban development, land monitoring, scene interpretation, and resource exploration [3]–[6]. Among these applications, HSI classification (HIC) is a common technique that enables analysis of the ground feature characteristics, thereby facilitating HSI applications.

A. Related Works

Many alternative methods have been developed for implementing HIC. For instance, people used pixels as the basic unit to classify HSIs and compared the similarity between pixels. Among these methods, the k-nearest neighbor (KNN) [7] method is widely used because of its simplicity and comprehensibility. However, this simple and crude method has high computational complexity and cannot mine the features of hyperspectral data. To extract features from HSIs, pattern recognition methods, such as support vector machine (SVM) [8], have been proposed to adapt to the nonlinear structure of hyperspectral data, while the SVM cannot solve the problem of multiclassification. Some attempts have been conducted to capture spatial features from the original data by extracting attribute profiles [9] and texture features [10], but the features extracted from 2-D images are not necessarily the spatial features of 3-D images.

With the development of artificial intelligence and neural networks, new learning-based methods, such as active learning [11], semisupervised learning [12], and artificial neural networks [13], have been exploited to classify tasks. Among these, convolutional neural networks (CNNs) [14] and recurrent neural networks (RNNs) [15] are widely used in HIC. Deep convolutional networks have benefited from the characteristics of convolution operators, including the use of local connections to reduce parameters and maintain the translation invariance of data and the high abstraction of deep networks;

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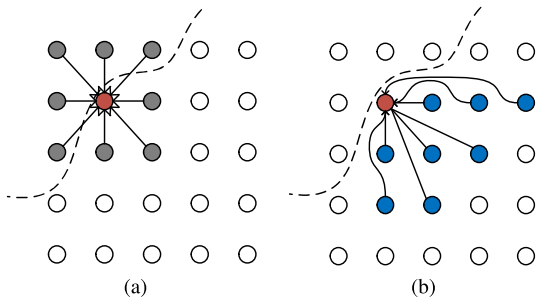


Fig. 1. Comparison of convolution. (a) 2-D-CNN convolution. (b) Graph convolution.

these networks have demonstrated a strong feature extraction ability on hyperspectral data and have achieved very competitive classification performance. In 2015, Hu *et al.* [16] and Mei *et al.* [17] first introduced 1-D-CNNs into HIC to extract spectral features. 2-D-CNNs [18]–[20] and 3-D-CNNs [21], [22] were then gradually proposed to adapt to the HSI features. Considering that HSIs are sequence-based in nature, to avoid losing sequence information, Mou *et al.* [23] used an RNN to classify HSIs. Hang *et al.* [24] proposed a cascaded RNN model to improve the feature recognition ability. Zhang *et al.* [25] introduced the local space sequence (LSS) method into RNNs to extract local and semantic information from HSIs.

Although the HIC methods based on convolutional networks achieve good results for classification accuracy, they face the following problem. This stems from the fact that the convolutional networks use a fixed shape kernel around the central pixel to extract features, which blurs the classification boundary and results in poor performance at the class boundary in HIC, as shown in Fig. 1(a). To overcome this shortage, the latest graph convolution networks (GCNs) are devoted to redefining convolution operators. From the spatial domain, as shown in Fig. 1(b), the graph convolution operator can provide local feature abstraction functions, such as the convolution kernel. The form of the graph convolution is more flexible and has greater potential for HSI with pixel-level classification. In [26], GCN was first applied to HIC, which explicitly uses adjacent nodes in the graph to approximate convolution. After that, superpixel segmentation [27], potential spatial similarity [28], and dynamic graph [29] were applied in GCN for HIC. Motivated by convolutional networks, Li *et al.* [30] integrated the success aspects of deep CNNs into GCNs, namely, residual connections [31], dense connections [32], and dilated convolutions [33], showing that GCNs can extract multiscale, deep-level abstract features.

B. Motivations and Contributions

In recent years, with the rapid development of spectral imaging technology, the resolution of HSIs has been improved, more information has been stored, and the number of bands has been considerably increased [34], [35]. This generates more complex spectral structure information, which, therefore, increases the pressure to achieve accurate classification of HSIs. Recent advances in deep learning-based solutions have provided powerful tools for HIC, while the performance of these methods is limited when encountering boundary problems and complex data [36]. Unfortunately, this is often the case considering a large number of HSI bands and spectral data redundancy. Currently, the spectral resolution of imaging spectrometers has reached the nanoscale. Moreover, the method for splitting light increases the complexity; in other words, the spectroscopic model is highly diversified, which adds a number of spectral and spatial resolutions to HSIs.

The improved resolution creates high-quality HSI, while it inevitably introduces two problems for HIC: the complex and redundant correlation between bands and increased information storage. For the first problem, the improvement in spectral resolution results in a larger number of bands, and the correlation between bands is enhanced. Intuitively, this hinders the effective interpretation of HSIs [37]. For example, when the vegetation reflection characteristics in the visible light band are low and the similarity in the visible spectrum region is large, the influence of terrain shadow on the solar reflection band is similar. As a result, the solar reflection region bands are severely interrelated. In these cases, it is challenging to compare the spectral correlation between HSI pixels when a large number of bands are present. As such, it is challenging to classify pixels with high correlation into the same category. For the second problem, the improvement in resolution leads to more complex information being stored in each unit area, which significantly increases the difficulty of HSI feature extraction. For example, the spectral characteristics of rocks are affected by the composition, structural color, and surface state, while the spectral characteristics of soil are mainly affected by the water content, organic matter, and other factors. The increased data indicate that more features are included; thus, feature extraction becomes more difficult. In summary, in response to the improved resolution of hyperspectral imagery, performing accurate and reliable HIC faces the following challenging issues.

1) *How to distinguish similar bands from high-dimensional spectral space when HSIs have considerable redundant information and high correlation between bands.* HSI bands are narrow, and the number of bands reaches hundreds. In addition, there is a high level of redundancy between the bands. For each surface feature, the causes of band correlation are manifold and involve the influence of light, moisture, and organic matter. Traditional methods, for example, distance-, projection-, and information measure-based similarity measurement methods, only focus on one aspect (e.g., spectral shape and amplitude). In addition to the characteristics of the spectra, the spatial distance will also affect the similarity between the spectra.
2) *How to focus on the spectral information that has the most influence from the HSI data with more redundancy.* There is considerable redundancy between HSIs, and the spectra interfere with each other. Due to the influence of the surrounding environment, such as different illumination angles, water content differences, and interference of radioactive substances, the corresponding spectral curves of the same surface feature may be different, and different surface features will present the same spectral curve. The same spectral curve features corresponding

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to these different features are part of the spectral redundancy, which will affect the classification effect to a certain extent.

3) *How to extract non-Euclidean features (e.g., physical and chemical structures) from hyperspectral data more efficiently.* Each pixel of an HSI represents a surface feature, and the ground features, such as physical structure and chemical composition, are non-Euclidean structures. At present, most of the deep learning methods, such as CNNs, are based on Euclidean structures and use fixed convolution operators to extract features. Some researchers classify HSIs based on the advantages of GCN in non-Euclidean structures. However, these studies only use a shallow GCN model to extract features and cannot mine deep features of HSIs.

To address the abovementioned challenges, we propose the deep attention graph convolutional network (DAGCN) to deal with the HIC by focusing on the problems stemming from the increasing resolution of HSIs. Specifically, considering that there are many bands in hyperspectral data and that the correlation between bands is complex, we design a new similarity measurement method, kernel spectral angle mapper, and spectral information divergence (KSAM-SID), for the proposed DAGCN to effectively cluster similar spectra. The standard KSAM [38] mainly concerns the difference in spectral shape in high-dimensional space. Given the considerable redundancy in multiple HSI bands, if only KSAM is used to calculate spectral similarity, the redundant spectral information will be included, which will degrade the efficiency of HIC. To solve this problem, we integrate KSAM with the weight SID [39] nonlinearly to constrain it spatially. By doing so, similar spectra are obtained, and different spectra are more obvious, thereby enabling reliable HIC. Second, considering the redundancy between spectra, HIC will be disturbed. We design an attention mechanism [40] to focus on the spectral information that has the greatest impact on classification. Finally, due to the complexity of hyperspectral data, many data are non-Euclidean structures. The proposed DAGCN introduces a deep graph convolutional network (DeepGCN) to mine the internal relationship between HSI data, with the aim of extracting non-Euclidean features from hyperspectral data more efficiently. The main contributions of this article are outlined as follows.

1) We design a new similarity measurement method, KSAM-SID, by exploring the correlation between the spectra of HSIs, which offers a more suitable method for capturing spectral characteristics and accurately aggregating HSI features.
2) We propose an appropriate attention graph structure generation method for the problem of graph convolution in HIC. Within the proposed attention graph structure, the nodes in the graph are most likely to belong to the same category, which can inhibit the effect of samples outside the category and, therefore, enhance the influence of samples within the category.
3) We design DeepGCNs to mine HSI features, which provides a new method for feature extraction in HIC. We adapt the dense connections and dilated convolutions into DeepGCNs to solve the oversmoothing and multiscale problems. In addition, we design a dynamic update mechanism to dynamically change the neighbor nodes to learn more effective graph structure features. A number of experiments based on well-known data sets validate the effectiveness of the proposed DAGCN method.

The remainder of this article is organized as follows. Section II presents the preliminary work. Section III details the proposed method for HSI. In Section IV, the data sets are first introduced, and then, the experimental setup, comparison, and analysis results are discussed. Next, the performance of the proposed method is statistically evaluated. Finally, Section V concludes this article.

II. Preliminaries

First, we define the basic symbols used in this article. Given the input HSI $H_{r\times c}^{\text{band}}$ (r, c, and band are the width, length, and number of image bands, respectively), we reshape the image $H_{r\times c}^{\text{band}}$ to H_N^{band} (hereinafter, we refer to it as H), where $N = r \times c$ represents the number of pixels. We define each spectral vector in H as $\boldsymbol{h}_i = \{h_i^1, h_i^2, \ldots, h_i^{\text{band}}\}(i \in [1, N])$.

A. Kernel Spectral Angle Mapper

The kernel spectral angle mapper (KSAM) [38] introduces the kernel technique, which implicitly maps the input to the high-dimensional feature space, which can better solve the nonlinear classification problem.

The basic idea is to map the spectra $\boldsymbol{h}_i$ and $\boldsymbol{h}_j$ to the high-dimensional feature space and then calculate the angle between the two spectral vectors in the high-dimensional space and finally measure the similarity d_{KSAM} according to the angle

$$d_{\text{KSAM}}(\boldsymbol{h}_i, \boldsymbol{h}_j) = \cos^{-1}\left(\frac{\Phi(\boldsymbol{h}_i, \boldsymbol{h}_j)}{\Phi(\boldsymbol{h}_i, \boldsymbol{h}_i) \cdot \Phi(\boldsymbol{h}_j, \boldsymbol{h}_j)}\right) \quad (1)$$

$$\Phi(\boldsymbol{h}_i, \boldsymbol{h}_j) = \exp\left(-||\boldsymbol{h}_i - \boldsymbol{h}_j||^2/(2\delta^2)\right) \quad (2)$$

where $d_{\text{KSAM}} \in [0, (\pi/2)]$. The larger the value of d_{KSAM}, the greater the gap between the two spectral vectors, and the smaller the similarity.

B. Spectral Information Divergence

The spectral information divergence (SID) [39] regards each spectral vector as information entropy with probability and statistical characteristics and treats the redundancy between the two spectral vector probabilities as the similarity.

Given a spectral vector $\boldsymbol{h}_i$, its statistical probability vector can be expressed as

$$\begin{aligned}\boldsymbol{p}_i &= \left\{p_i^1, p_i^2, \ldots, p_i^{\text{band}}\right\} \\ &= \left\{h_i^1/h_i^{\text{sum}}, h_i^2/h_i^{\text{sum}}, \ldots, h_i^{\text{band}}/h_i^{\text{sum}}\right\}\end{aligned} \quad (3)$$

where $h_i^{\text{sum}} = \sum_{j=1}^{\text{band}} h_i^j$ represents the sum of spectral features.

The relative entropy (also known as KL divergence) of the probability statistics vector $\boldsymbol{h}_i$ and $\boldsymbol{h}_j$ can be calculated as

$$d(\boldsymbol{h}_i||\boldsymbol{h}_j) = \sum_{k=1}^{\text{band}} h_i^k \ln \frac{h_i^k}{h_j^k}. \quad (4)$$

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The relative entropy between the spectra does not satisfy symmetry, that is, $d(\boldsymbol{h}_i||\boldsymbol{h}_j) \neq d(\boldsymbol{h}_j||\boldsymbol{h}_i)$. According to the cross entropy between spectra, the SID can be obtained

$$d_{\mathrm{SID}}(\boldsymbol{h}_i||\boldsymbol{h}_j) = d(\boldsymbol{h}_i||\boldsymbol{h}_j) + d(\boldsymbol{h}_j||\boldsymbol{h}_i). \tag{5}$$

The larger the value of d_{SID} is, the lower the similarity between spectra and the higher the similarity.

C. Graph Convolution of Hyperspectral Data

In graph signal processing, the Laplace operator is an operator reflecting local smoothness and can describe the signal difference between a central node and a neighboring node. By normalizing the Laplacian matrix, a more robust graph-structured data representation can be obtained, which has the property of real symmetric positive semidefinite

$$L = I_N - D^{-1/2} A D^{-1/2}, \quad D_{ii} = \sum_j A_{ij} \tag{6}$$

where A is the adjacency matrix of the graph and D is a node degree matrix that records the degree of each node.

Given the input HSI H, the spectral convolution [41] is defined as follows:

$$g_\omega \star \mathrm{H} = U g_\omega U^T \mathrm{H} \tag{7}$$

where g_ω is the filter with parameter ω, U is the matrix composed of the eigenvector of the normalized graph Laplacian $L = U\Lambda U^T$, and Λ is the diagonal matrix composed of the eigenvalues of L. We can think of g_ω as a function of the eigenvalue of L, that is, $g_\omega(\omega)$. To avoid the high cost of feature decomposition, Hammond *et al*. [42] approximated $g_\omega(\Lambda)$ using the Chebyshev polynomial

$$g_\omega(\Lambda) \approx \sum_{k=0}^{K} \omega_k' T_k(\tilde{\Lambda}) \tag{8}$$

where $\tilde{\Lambda} = 2\Lambda/\lambda_{\max} - I_N$, $\lambda_{\max}$ is approximately 2, and then, $\tilde{\Lambda} = \Lambda - I_N$. Accordingly

$$g_\omega \star \mathrm{H} \approx \sum_{k=0}^{K} \omega_k' T_k(\tilde{L}) \tag{9}$$

where $\tilde{L} \approx L - I_N$. We set $K = 1$ as proposed by Kipf and Welling [41]; then

$$\begin{aligned} g_\omega \star \mathrm{H} &\approx \omega_0' \mathrm{H} + \omega_1'(L - I_N)\mathrm{H} \\ &= \omega_0' \mathrm{H} - \omega_1' D^{-1/2} A D^{-1/2} \mathrm{H}. \end{aligned} \tag{10}$$

Let $\Omega = [\omega_0', -\omega_1']$; then

$$g_\omega \star \mathrm{H} \approx \left(I_N + D^{-1/2} A D^{-1/2}\right)\mathrm{H}\Omega. \tag{11}$$

The eigenvalues of $I_N + D^{-1/2} A D^{-1/2}$ in (11) are in the range of [0, 2]. The continuous use of this operator in deep neural networks will lead to numerical instability and gradient explosion/disappearance problems. To solve these problems, the operator is improved to be $\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}$, where $\tilde{A} = A + I_N$. This not only alleviates the network training problem but also considers the node self-loop

$$g_\omega \star \mathrm{H} = \tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}\mathrm{H}\Omega. \tag{12}$$

Thus, the space GCN is expressed as

$$\begin{aligned} \mathrm{H}^i &= F^i\left(\mathrm{H}^{i-1}, W^{i-1}\right) \\ &= \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}\mathrm{H}^{i-1}\Omega^{i-1}\right) \end{aligned} \tag{13}$$

where H^i represents the characteristic matrix of the ith layer, F represents the graph convolution operation, Ω^i represents the weight matrix of the ith layer, and σ represents the activation function, such as ReLU.

III. Proposed Model

In this section, we introduce the HIC process in detail, and its framework is shown in Fig. 2. Given the original image, we use the KSAM-SID similarity measurement method to obtain the similarity matrix. Then, we construct the attention KSAM-SID graph structure and use the generated graph structure for graph convolution. By doing so, the features of the image are obtained using deep attention GCN (DAGCN), and the graph structure of each layer is dynamically updated to adapt to the change in the GCN middle layer. The number of layers of the network is 10, including one layer of graph convolution and three layers of the same dense block. Finally, we use the fully connected layer to classify the features of the DAGCN output.

This section includes the following contents: a new similarity measurement method KSAM-SID (see Section III-A), construction of the attention KSAM-SID graph structure (see Section III-B), and the structure of the DeepGCNs (see Section III-C).

A. KSAM-SID Similarity Measurement

KSAM focuses on the difference in spectral shape in high-dimensional space. However, some bands of HSIs have redundancy. If only KSAM is used to calculate spectral similarity, redundant spectral information will be included. Therefore, it is necessary to combine the information entropy of SID to constrain KSAM.

In this article, KSAM is used to weight SID nonlinearly [43]. This takes full advantage of the advantages of SID and KSAM in spectral similarity calculations so that similar spectra are more similar and spectral differences are more obvious. According to the formula, the mixed measure is given as follows:

$$d_{\mathrm{mix}}(\boldsymbol{h}_i, \boldsymbol{h}_j) = d_{\mathrm{SID}}(\boldsymbol{h}_i||\boldsymbol{h}_j) \times \sin\left(d_{\mathrm{KSAM}}(\boldsymbol{h}_i, \boldsymbol{h}_j)\right) \tag{14}$$

where $\sin(d_{\mathrm{KSAM}}(\boldsymbol{h}_i, \boldsymbol{h}_j))$ is to constrain d_{KSAM} to [0,1]. d_{SID} adapts to the undirected graph convolution graph structure.

In addition, due to the great probability that the spectra closer to each other belong to the same class and the spectra with a longer distance are not of the same category, it is necessary to introduce spatial constraints. The Chebyshev distance is used to constrain the spectral mixing measure

$$\tilde{d}_{\mathrm{mix}}(\boldsymbol{h}_i, \boldsymbol{h}_j) = \log(d_{ij}) \times d_{\mathrm{mix}}(\boldsymbol{h}_i, \boldsymbol{h}_j) \tag{15}$$

where $d_{ij} = \max(|i_1 - i_2|, |j_1 - j_2|)$, and $(\cdot)_1$ and $(\cdot)_2$, respectively, represent the horizontal and vertical coordinates of the pixel.

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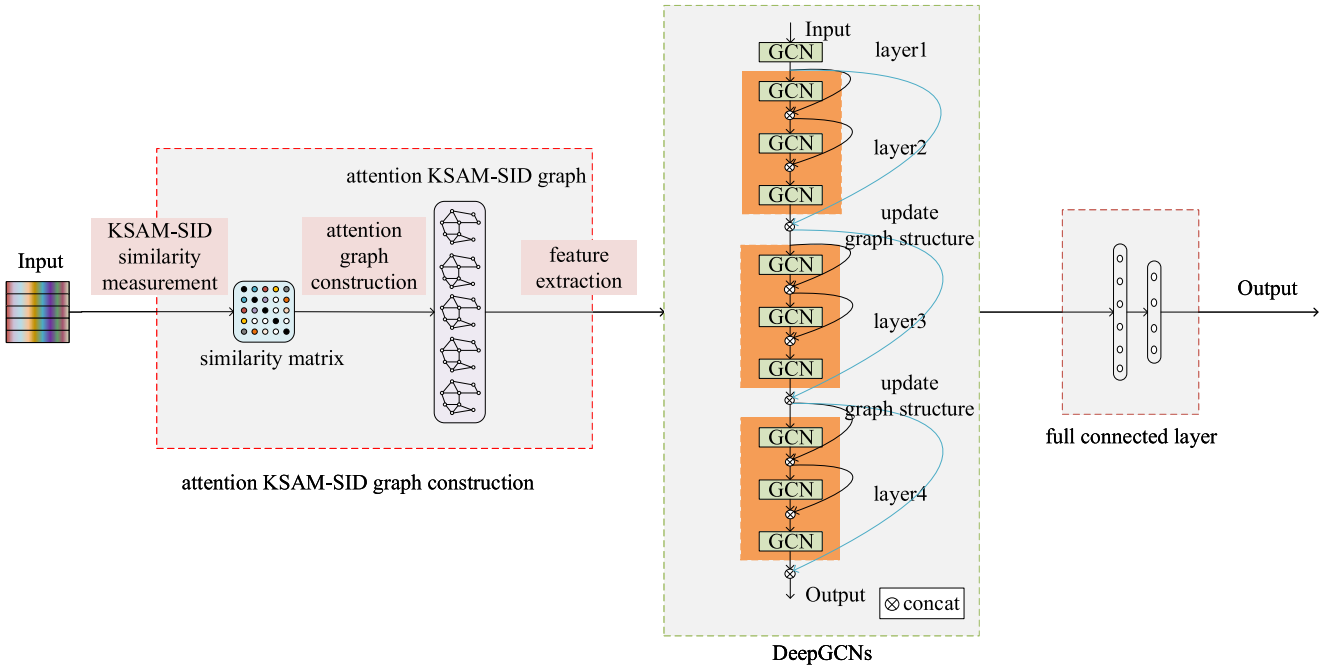


Fig. 2. Classification framework of HSIs based on ten layers of the DAGCN. First, we use the KSAM-SID similarity measurement method to obtain the similarity matrix. Then, we construct the attention KSAM-SID graph structure and use the generated graph structure for graph convolution. In this article, we use DeepGCNs to extract features. The number of layers of the network is 10, including one layer of graph convolution and three layers of the same dense block. Finally, the classification is carried out through the fully connected layer.

B. Attention KSAM-SID Graph Structure

The current graph convolutional network does not have a suitable method for generating graph structures on HSIs. We interpret HSIs as graph structures based on spectral similarity and design the attention KSAM-SID graph adjacency matrix to construct the graph structure.

The attention KSAM-SID graph adjacency matrix process is mainly divided into three stages, as shown in Fig. 3.

1) First Stage: Given a spectral vector $\boldsymbol{h}_i$, we calculate the KSAM-SID similarity between it and all spectra H

$$\tilde{d}_{\text{mix}}(\boldsymbol{h}_i, \text{H}) = \left[\tilde{d}_{\text{mix}}(\boldsymbol{h}_i, \boldsymbol{h}_1), \tilde{d}_{\text{mix}}(\boldsymbol{h}_i, \boldsymbol{h}_2), \ldots, \tilde{d}_{\text{mix}}(\boldsymbol{h}_i, \boldsymbol{h}_N)\right]. \tag{16}$$

The similarity matrix of the whole HSI S can be calculated according to the following formula:

$$S = \begin{bmatrix} \tilde{d}_{\text{mix}}(\boldsymbol{h}_1, \boldsymbol{h}_1) & \tilde{d}_{\text{mix}}(\boldsymbol{h}_1, \boldsymbol{h}_2) & \cdots & \tilde{d}_{\text{mix}}(\boldsymbol{h}_1, \boldsymbol{h}_N) \\ \tilde{d}_{\text{mix}}(\boldsymbol{h}_2, \boldsymbol{h}_1) & \tilde{d}_{\text{mix}}(\boldsymbol{h}_2, \boldsymbol{h}_2) & \cdots & \tilde{d}_{\text{mix}}(\boldsymbol{h}_2, \boldsymbol{h}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{d}_{\text{mix}}(\boldsymbol{h}_N, \boldsymbol{h}_1) & \tilde{d}_{\text{mix}}(\boldsymbol{h}_N, \boldsymbol{h}_2) & \cdots & \tilde{d}_{\text{mix}}(\boldsymbol{h}_N, \boldsymbol{h}_N) \end{bmatrix}. \tag{17}$$

For each spectrum vector, its own similarity distance is 0, that is, $\tilde{d}_{\text{mix}}(\boldsymbol{h}_i, \boldsymbol{h}_i) = 0$. S is a symmetric matrix.

The nearest neighbors of each sample are determined according to the similarity value. To avoid the influence of distant samples on the sample characteristics in the propagation process, the KNN algorithm is used to select the nearest neighbors of the samples. For each spectral vector $\boldsymbol{h}_i$, we choose the k most similar spectra as neighbors $\Theta_i = \{\boldsymbol{h}_{t_1}, \boldsymbol{h}_{t_2}, \ldots, \boldsymbol{h}_{t_k}\}, t_1, t_2, \ldots, t_k \in [1, N]$. We keep the part of $S(\boldsymbol{h}_i, \Theta_i)$ and set the others to 0.

2) Second Stage: Note that the more similar the spectral vector is, the smaller the similarity distance is (the similarity distance of itself is 0), and the influence on feature aggregation is smaller. For better feature aggregation, we transform the similarity matrix as follows:

$$\hat{S}_{i\cdot} = \begin{cases} \min(S_{i\cdot}) * \dfrac{1}{S_{i\cdot}}, & S_{i\cdot} \neq 0 \\ 1, & S_{i\cdot} = 0. \end{cases} \tag{18}$$

Because the similarity distance between a spectral vector and other spectral vectors is often on the order of magnitude, to prevent the data from being too large or too small, we add the $\min(S_{i\cdot})$ coefficient to normalize the similarity value.

The similarity matrix $\hat{S}$ is normalized by row

$$\ddot{S}_i = \text{softmax}\left(\hat{S}_i\right). \tag{19}$$

3) Third Stage: The purpose of this article is to use the similarity between neighborhood nodes to aggregate them, but the current $\ddot{S}$ matrix does not contain the similarity between its own samples. Add the appropriate values to the diagonal elements of the similarity matrix, and then, softmax normalization is carried out

$$\tilde{S} = \text{softmax}\left(\ddot{S} + I_N\right), \quad \sum_{j=1}^{N} \tilde{S}_{ij} = 1, i \in [1, N]. \tag{20}$$

After the above operations, the graph is a directed graph. We need to adjust the graph and set the weight between some nodes to 0 so that the graph becomes an undirected graph. The similarity matrix $\tilde{S}$ is regarded as the attention graph adjacency matrix, which is different from the adjacency matrix. Since $\sum_{j=1}^{N} \tilde{S}_{ij} = 1$ and $Dii = \sum_j A_{ij}$, the space GCN [see (13)]

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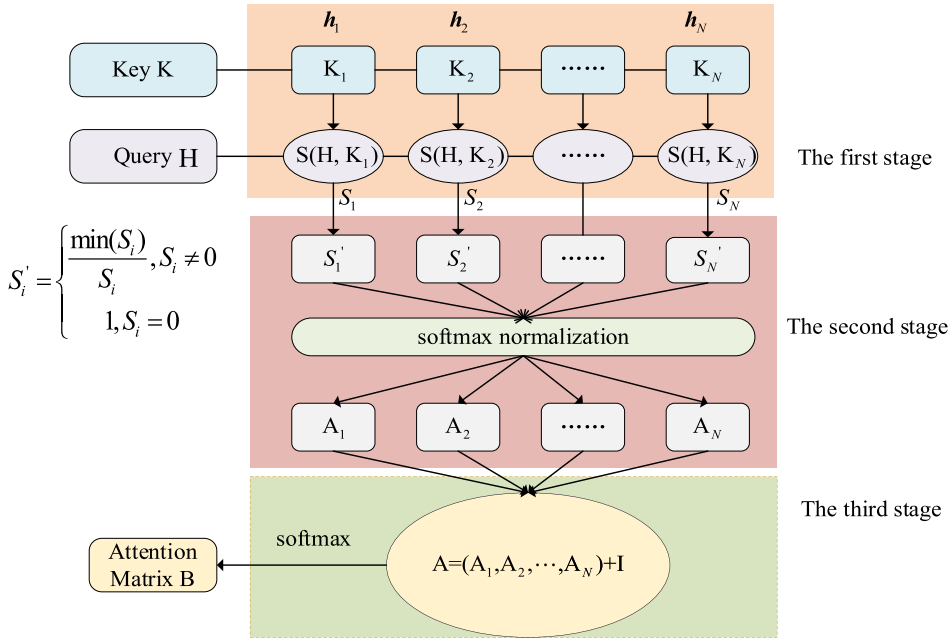


Fig. 3. Schematic of the attention KSAM-SID graph adjacency matrix. Key is the spectral vector to be compared, the query is all spectral vectors, and $S(H, K_i)$ is the KSAM-SID measurement result. S_i is the similarity vector after the KNN operation. B is the result of the attention KSAM-SID graph adjacency matrix.

can be simplified to

$$\mathrm{H}^{i}=\sigma\left(\tilde{S}\mathrm{H}^{i-1}\Omega^{i-1}\right). \tag{21}$$

C. DeepGCNs

The graph convolution operation can smooth the signal and, therefore, helps to mine the inner relationship between the neighborhood nodes. Note that, with the increase in the number of GCN layers, the representation node vectors tend to be consistent, and the node discrimination will become increasingly worse [44], [45]. As a result, the characteristics of the graph vertices converge to the same value, which inevitably deteriorates the performance of the node classification task. As a solution, motivated by Li *et al.* [30], we utilize dilated convolutions and dense connections to solve this problem. In addition, after a series of graph convolution operations, the graph feature is not immutable, so we design a dynamic update graph structure module to adapt to the graph feature.

1) Dense Connections Module: According to DenseNet [32], the efficient utilization of interlayer information flow in the GCN is shown as follows:

$$\begin{aligned} H^{i+1} &= F^{i+1}\left(H^{i}, \Omega^{i}\right) \\ &= T\left(F^{i}\left(H^{i}, \Omega^{i}\right), H^{i}\right) \\ &= T\left(F^{i}\left(H^{i}, \Omega^{i}\right), \ldots, F^{0}\left(H^{0}, \Omega^{0}\right), H^{i}\right). \end{aligned} \tag{22}$$

The operator T is a point-level cascade function that can fuse the output characteristic graph H^0 with the intermediate outputs of all GCNs.

With the number of layers increasing, the feature dimension of H^i increases, which results in a large increment of the runtime and memory requirements. Therefore, we design the dense block module to reduce the computing cost. The dense block module is shown in Fig. 4.

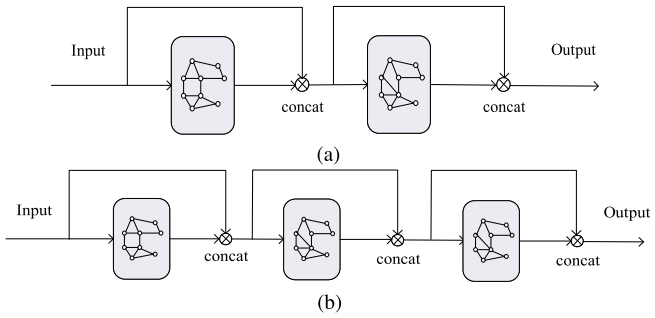


Fig. 4. Dense block. (a) Dense block 2 layers. (b) Dense block 3 layers.

2) Dilated Convolutions Module: In the field of remote sensing, image structures are different, the image scale span is large, and the homogeneous regions in the same image are different. Combining dilated convolution into DenseGCNs can effectively extract both high-order abstract features and multiscale features.

In this study, after each dense block, dilated convolution is used to find the extended neighborhood and establish the DilatedGCN. Specifically, the DilatedGCN graph $G = (V, E)$ with hole ratio d is used to take a sample of every d neighbors in the $k \times d$ neighborhood

$$\mathrm{Nei}^{(d)}(v) = \left\{v_1, v_{1+d}, v_{2+d}, \ldots, v_{1+(k-1)d}\right\} \tag{23}$$

where $\mathrm{Nei}^{(d)}(v)$ denotes the d hole neighborhood and represents the result of sampling in the $k \times d$ neighborhood $\{v_1, v_2, \ldots, v_{k\times d}\}$ of the vertex. By applying the KNNs, a suitable distance measure standard is generated. The proposed DilatedGCN method is, therefore, suitable for the spectral similarity used in hyperspectral data. Fig. 5 shows the schematic of DilatedGCN when d is 1, 2, and 3.

3) Dynamic Graph Representation: Most GCNs have a fixed graph structure and only update vertex features in each

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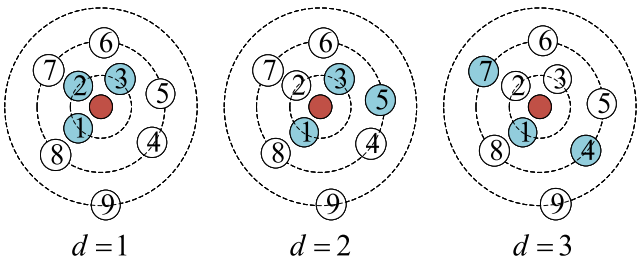


Fig. 5. Schematic of DilatedGCN.

iteration. Using dynamic graph convolution to change the graph structure at each layer and dynamically change the neighbor nodes can effectively alleviate the oversmoothing problem and generate a larger receptive field [46], [47]; this allows the network to learn more effective graph features.

For these reasons, we recalculate the graph adjacency matrix according to the output characteristics after each dense block. In (21), the size of $\tilde{S}$ is $[N, N]$, the size of H^{i-1} is $[N, h_1]$, and the size of Ω^{i-1} is $[h_1, h_2]$, where h_1 and h_2 are the number of hidden units. After calculation, the output size is $[N, h_2]$. Therefore, the size of the graph adjacency matrix is still $[N, N]$. We only need to update the graph adjacency matrix on the output H^i.

We use the cosine distance to update the graph structure dynamically. Since we normalize the GCN of each layer with L2 normalization, the cosine similarity is the inner product of the graph feature; that is, $H^l(H^l)^T$. The calculation formula of the similarity matrix $\tilde{S}^l$ of the l layer is

$$\tilde{S}^l = \text{softmax}\left(H^l\left(H^l\right)^T\right) + I_N \tag{24}$$

where I_N joins the similarity of its own nodes, which can effectively prevent the large fluctuation of the training process.

Then, we use the KNN algorithm to obtain neighbor nodes according to the hole ratio d and normalize the values between the neighbors. Finally, we set the other values in the graph adjacency matrix to 0.

IV. Experiments and Results

A. Data Sets

In this section, we conduct experiments based on three classical hyperspectral data sets, namely, Indian Pines, Pavia University, and Salinas, and a challenging data set Houston, to evaluate the performance of the proposed method.

AVIRIS is an airborne visible/infrared imaging spectrometer belonging to the U.S. Jet Propulsion Laboratory. The AVIRIS imaging spectrometer has an imaging wavelength range of 0.4–2.5 μm and can produce images of ground objects in 224 continuous wavebands. In June 1992, AVIRIS imaged an Indian pine forest in Indiana with a spatial resolution of 20 m/pixel. The range of 145×145 pixels is intercepted to obtain the Indian pine image. Finally, 200 bands of the Indian Pines data set are obtained by denoising 24 bands covering the water absorption area. In addition to the background, there are 10 249 marked pixels in the data set, including 16 categories.

ROSIS is a reflection optical system imaging spectrometer from the German National Aeronautics and Space Administration. The wavelength range is 0.43–20.86 μm. It can continuously produce images of ground objects in 115 continuous wavebands. In 2003, ROSIS imaged the city of Pavia in Italy with a spatial resolution of 1.3 m/pixel and used a pixel size of 610×340 to obtain the image of Pavia University. The result is a 103-band Pavia University data set obtained by denoising 12 bands covering the water absorption area. In addition to the background, there are 42 776 marked pixels and a total of nine categories.

Salinas was captured by a 224-band AVIRIS sensor in Salinas Valley, CA, USA, with a spatial resolution of 3.7 m/pixel and a captured pixel size of 512×217. Similar to the Indian Pines scene, 20 water-absorbing bands were discarded, and 204 bands were retained. In addition to the background, there are 54 129 marker pixels in the data set, including 16 categories.

The data set Houston was collected by NCALM at the University of Houston (UH) in June 2012, covering the UH campus. The data were provided by the 2013 IEEE GRSS Data Fusion Competition. The data size is 349×1905, including 144 spectral bands ranging from 364 to 1046 nm, with a total of 15 categories.

B. Experimental Setup

Our experiment is carried out on a computer with a 3.20-GHz Intel Core i5-6500 CPU with 8 GB of RAM and a GTX 1070 Ti GPU.

We use the general model evaluation criteria, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficients (Kappa), on the standard data sets to verify the effectiveness of the proposed method. Approximately, 10% of the samples are randomly selected as the training set for Indian Pines and Houston, 6% of the samples are randomly selected as the training set for Pavia University, and 5% of the samples are randomly selected as the training set for Salinas. For categories with fewer samples, at least ten samples are randomly selected for the training set, while other samples are used as the test set.

The numbers of training and test samples for each category in the four hyperspectral data sets are shown in Tables I–IV, respectively.

1) AttentionGCN: We name the two-layer attention GCN as AttentionGCN.

In order to verify the effectiveness of our proposed KSAM-SID method, we compare the common similarity measurement methods, L2 distance, SAM [48], SID [39], and KSAM [38] with our method, and the parameters of each experiment are consistent.

The graph structure of graph convolution greatly influences feature extraction for HSIs, and the graph structure is determined by node neighbors. To obtain the best results, we use a different number of k-nearest spatial neighbors to compare the classification results of different graph structures for AttentionGCN. We gradually increase the value of the KNNs to conduct experiments. The values for the KNNs are 8, 16, 24, 32, 40, 48, 56, 64, 72, and 80 for comparison.

The hidden units of the two-layer GCN network are [300, 200]. The learning rate of the Adam optimization algorithm is set to 0.01, the number of iterations is 500, attenuation rates

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TABLE I
NUMBER OF TRAINING AND TEST SAMPLES ON THE INDIAN PINES DATA SET

Class No.	Class Name	Training	Test	TOTAL
1	Alfalfa	10	36	46
2	Corn-notill	143	1285	1428
3	Corn-min	83	747	830
4	Corn	24	213	237
5	Grass/Pasture	49	434	483
6	Grass/Trees	73	657	730
7	Grass/Pasture-mowed	10	18	28
8	Hay-windrowed	48	430	478
9	Oats	10	10	20
10	Soybeans-notill	98	874	972
11	Soybeans-min	246	2209	2455
12	Soybean-clean	60	533	593
13	Wheat	21	184	205
14	Woods	127	1138	1265
15	Building-Grass-Trees-Drives	39	347	386
16	Stone-steel Towers	10	83	93
	TOTAL	1051	9198	10249

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES ON THE PAVIA UNIVERSITY DATA SET

Class No.	Class Name	Training	Test	TOTAL
1	Asphalt	398	6233	6631
2	Meadows	1119	17530	18649
3	Gravel	126	1973	2099
4	Trees	184	2880	3064
5	Painted metal sheets	81	1264	1345
6	Bare Soil	302	4727	5029
7	Bitumen	80	1250	1330
8	Self-Blocking Bricks	221	3461	3682
9	Shadows	57	890	6631
	TOTAL	2568	40208	42776

λ are set to 5e-6, and dropout parameters are set to 0.5 (to prevent overfitting).

2) Comparative Experiment: We selected several advanced methods for comparison, including SVM [8], SAE-LR [49], EPF [50], SVM-3DG [51], 2-D-CNN [21], 3-D-CNN [22], GCN [41], FuNet-C [52], MDGCN [27], and our proposed DAGCN.

All the experimental results are averaged after five runs. The experiments presented in this study have achieved competitive results under the premise of ensuring stability.

The process and parameters of DAGCN are set as follows. First, we use KSAM-SID to obtain the similarity matrix, and the parameter δ of KSAM is 0.55. Then, we build the graph structure. On the Indian Pines and Pavia University data sets, k is set to 24. On the Salinas data set, k is set to 16. After that, we use DeepGCNs to extract features. If the layer number of the dense block is 2, the hole ratio d is 1. If the layer number of the dense block is 3, the hole ratio d is 2. The hole ratio d increases with the number of layers of the dense block. Finally, we use two fully connected layers [256, nclass] to obtain the classification results, where nclass is the class number of the image.

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES ON THE SALINAS DATA SET

Class No.	Class Name	Training	Test	TOTAL
1	Brocoli-green-weeds-1	101	1908	2009
2	Brocoli-green-weeds-2	187	3539	3726
3	Fallow	99	1877	1976
4	Fallow-rough-plow	70	1324	1394
5	Fallow-smooth	134	2544	2678
6	Stubble	198	3761	3959
7	Celery	179	3400	3579
8	Grapes-untrained	564	10707	11271
9	Soil-vinyard-develop	311	5892	6203
10	Corn-senesced-green-weeds	164	3114	3278
11	Lettuce-romaine-4wk	54	1014	1068
12	Lettuce-romaine-5wk	97	1830	1927
13	Lettuce-romaine-6wk	46	870	916
14	Lettuce-romaine-7wk	54	1016	1070
15	Vinyard-untrained	364	6904	7268
16	Vinyard-vertical-trellis	91	1716	1807
	TOTAL	2713	51416	54129

TABLE IV
NUMBER OF TRAINING AND TEST SAMPLES ON THE HOUSTON DATA SET

Class No.	Class Name	Training	Test	TOTAL
1	Healthy grass	138	1374	1512
2	Stressed grass	146	1454	1600
3	Synthetic grass	80	795	875
4	Trees	127	1264	1391
5	Soil	130	1298	1428
6	Water	34	339	373
7	Residential	148	1476	1624
8	Commercial	136	1354	1490
9	Road	156	1554	1710
10	Highway	143	1424	1567
11	Railway	157	1566	1723
12	Parking Lot 1	143	1429	1572
13	Parking Lot 2	64	632	696
14	Tennis Court	52	513	565
15	Running Track	80	798	878
	TOTAL	1734	17270	19004

Here, DAGCN layers are set to 10. First, a GCN with 64 hidden layer units is implemented, and then, three dense blocks are used. The number of layers of the dense block is 3, and the number of hidden layer units is [64, 128, 192]. The network is updated iteratively with the Adam optimizer at a learning rate of 0.001, and the maximum number of iterations is 500. After each layer, batch normalization is used. The dropout of the fully connected layer is set to 0.3 to prevent overfitting.

C. Results of AttentionGCN

1) Results of Different Similarity Measurement Methods: We conduct experiments on different similarity measurement methods on the Indian Pines data set. The results are shown in Table V. We find that, although the simple similarity measurement methods have less computational overhead, they do not take full account of the spectral characteristics, resulting

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TABLE V

COMPARISON OF DIFFERENT SIMILARITY MEASURES ON INDIAN PINES DATA SET

	L2	SAM	SID	KSAM	KSAM-SID
OA	73.21	80.07	70.98	72.98	98.51
Time(s)	39.54	45.14	31.16	40.82	80.41

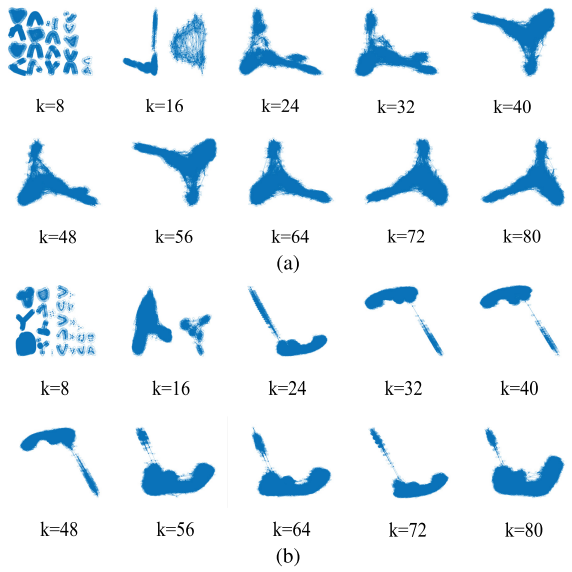


Fig. 6. Graph structure generated with different k's. (a) Graph structure of Indian Pines. (b) Graph structure of Pavia University.

in poor classification results. In contrast, our method not only considers the shape features of the spectrum and the features in high-dimensional space but also constrains the spatial position; hence, the OA has been significantly improved.

2) Visualization of Graph Structure: We visualize the graph structure generated under different k on the Indian Pines and Pavia University data sets, as shown in Fig. 6.

When the value of k is small, such as $k = 8$, the graph structure contains many independent subgraphs. With increasing k, the number of subgraphs decreases or even disappears. When $k = 16$, there are only two subgraphs, and there are no subgraphs after $k = 24$. When k reaches a certain value ($k = 24$), the shape of the graph structure remains unchanged but becomes dense by increasing the value of k.

3) Training Network: We compared the accuracy and loss of training under different k's, as shown in Fig. 7.

On the Indian Pines data set, we can clearly see that, with the increase in k, the network convergence is slower. For example, compared with the red and blue lines, the training accuracy of the red line increases faster, and the loss decreases faster. On the Pavia University data set, there is no significant difference in training accuracy and loss. On the Salinas data set, with the increase in k, the convergence value of the training accuracy decreases, and the training loss value increases.

4) Classification Results: We compare the results of OA, AA, and Kappa under different k conditions, as shown in Table VI.

We call the black part of Table VI the critical value $\tilde{k}$. When k gradually increases to $\tilde{k}$, OA, AA, and Kappa show an upward trend. When k reaches $\tilde{k}$, OA, AA, and Kappa

TABLE VI

CLASSIFICATION RESULTS OF DIFFERENT k'S

	Indian Pines			Pavia University			Salinas		
k	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
8	98.06	98.32	97.79	98.25	97.12	97.68	96.31	98.32	95.89
16	98.45	**98.90**	98.24	98.59	97.43	98.13	**96.95**	**98.58**	**96.61**
24	**98.51**	98.81	**98.30**	**99.22**	**98.85**	**98.97**	96.40	98.33	95.99
32	98.30	98.51	98.06	98.92	98.62	98.57	96.40	98.16	95.99
40	97.64	98.14	97.30	98.78	98.48	98.39	96.11	97.96	95.66
48	97.29	97.73	96.91	98.17	97.73	97.58	95.24	97.40	94.69
56	97.40	97.93	97.03	98.60	97.95	98.14	95.77	97.84	95.29
64	95.55	95.66	94.91	98.01	97.48	97.37	95.32	97.55	94.79
72	96.22	96.93	95.68	98.02	97.36	97.38	95.06	97.52	94.51
80	95.92	96.60	95.34	97.91	96.92	97.23	94.96	97.29	94.38

achieve the best effect. When k gradually increases after $\tilde{k}$, OA, AA, and Kappa show a downward trend. Interestingly, we find that the critical value of the graph structure in Fig. 6 is consistent with that in Table V, which means that the best result is obtained when the shape of the graph structure is no longer changed.

D. Layer Setting for the DAGCN

To study the influence of the network depth on the classification results, we conducted experiments on the Indian Pines data set every two layers. The total number of iterations is 500. The experimental results are shown in Fig. 8.

Based on the results from Fig. 8, we see that the accuracy rate increases gradually before layer 8, and then, the accuracy rate basically remains unchanged. As the number of layers increases, the computational complexity increases, and the process becomes increasingly time-consuming. In the case that the experimental results are not affected, we choose 10 as the number of layers.

E. Results of Comparative Experiments

1) Indian Pines Data Set: The classification results of each algorithm on the Indian Pines data set are shown in Fig. 9. It can be observed that our method has fewer spots in the classification map, and the classification is more accurate in the boundary area of different categories.

The specific classification results for each algorithm are shown in Table VII. We see that our method achieves the best classification effect in three evaluation indexes, namely, OA, AA, and Kappa. When the training samples are small, the accuracy of our method reaches 100% in categories 1, 7, 9, and 16, which proves that the flexible convolution form of the graph convolution is more conducive to the distinction of boundaries. Compared with 2-D-CNN and 3-D-CNN, our method improves the OA by 5.93%, which shows the advantages of GCN in HIC. Compared with the MDGCN, which has the best effect in comparison, our method improves the OA by 1.75%, which indicates the effectiveness of our KSAM-SID similarity measurement method and DeepGCNs.

Fig. 12 illustrates the high-dimensional features of hyperspectral data by T-SNE. The results in Fig. 12 (a)–(e) denote the T-SNE feature visualization of the Indian Pines data set.

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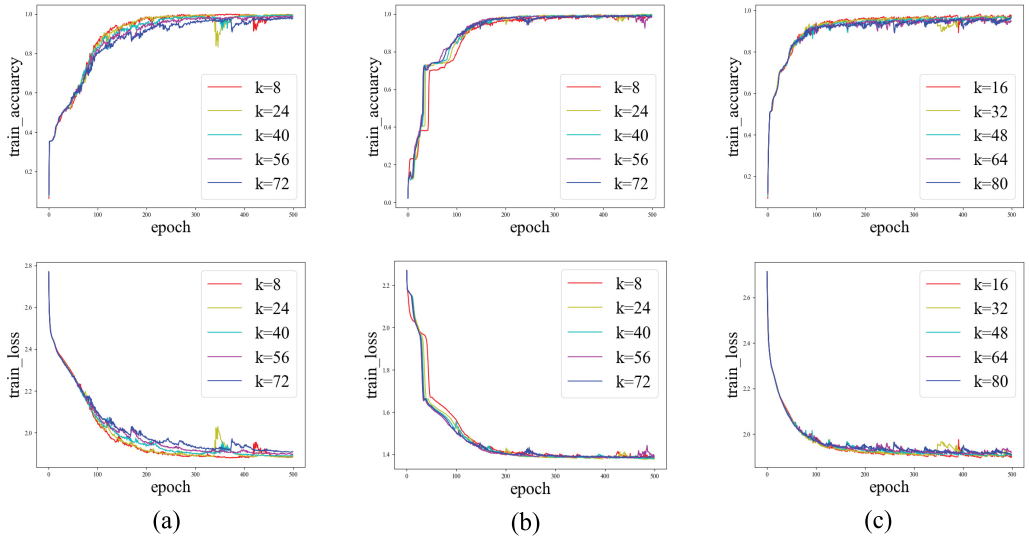


Fig. 7. Training accuracy and loss under different k's. (a) Indian Pines. (b) Pavia University. (c) Salinas.

TABLE VII
COMPARISON OF EXPERIMENTAL CLASSIFICATION RESULTS ON INDIAN PINES

Class No.	SVM	SAE-LR	EPF	SVM-3DG	2D-CNN	3D-CNN	GCN	FuNet-C	MDGCN	DAGCN
1	47.62±9.37	78.75±2.61	100±0.00	93.00±8.67	75.61±19.51	94.78±5.00	59.43±5.11	92.17±4.24	92.78±4.21	**100±0.00**
2	74.96±2.17	73.82±2.48	95.47±5.00	92.00±3.24	88.85±3.78	89.63±1.37	51.22±0.99	80.85±4.13	94.34±1.59	**99.13±0.17**
3	64.79±3.33	66.52±2.68	95.02±3.50	95.00±3.04	92.58±0.67	91.59±1.06	58.39±1.18	85.18±2.62	**96.68±2.34**	93.11±0.42
4	54.49±6.45	49.48±2.49	71.23±7.29	90.00±6.19	95.15±2.58	84.97±13.50	64.43±1.78	80.00±6.06	97.56±2.03	**99.25±0.26**
5	87.08±2.73	82.69±2.02	98.46±1.40	96.00±2.43	88.47±6.98	94.74±3.46	82.95±0.97	93.71±1.00	95.95±2.43	**100±0.00**
6	96.44±1.21	95.32±0.72	98.35±1.21	98.00±1.18	94.48±2.21	98.54±0.56	91.77±0.39	98.60±1.25	98.42±0.39	**99.70±0.00**
7	66.15±18.52	90.00±7.24	100±0.00	98.00±2.98	84.94±10.09	100±0.00	96.47±3.22	97.86±1.96	90.00±8.24	**100±0.00**
8	98.00±1.41	98.63±0.66	99.85±0.32	99.00±0.12	97.09±2.96	99.28±0.67	87.83±0.65	99.2±0.79	99.81±0.42	**100±0.00**
9	38.89±6.80	78.00±4.47	100±0.00	100.00±0.00	79.65±23.51	99.00±2.23	88.89±0.00	100±0.00	94.00±13.42	**100±0.00**
10	71.41±3.59	78.09±1.89	82.6±7.13	89.00±4.59	85.01±12.67	89.79±2.95	57.80±0.76	83.60±4.13	92.59±2.62	**99.13±0.07**
11	84.39±1.86	86.74±0.76	95.74±2.65	98.00±0.96	94.28±2.44	92.63±1.78	85.30±0.75	90.69±3.02	98.21±0.45	**98.42±0.06**
12	66.78±3.07	50.89±1.08	95.71±1.36	97.00±1.14	91.38±4.68	88.06±3.34	47.22±1.02	80.87±4.46	96.36±2.19	**98.23±0.10**
13	96.33±1.60	98.75±0.00	100±0.00	98.00±1.3	96.91±3.12	99.12±0.72	95.52±0.25	99.80±0.44	99.46±1.22	**100±0.00**
14	96.38±0.84	87.44±0.95	98.39±1.36	98.00±0.43	97.25±2.73	97.58±1.46	94.62±0.17	95.82±1.49	99.12±1.01	**100±0.00**
15	49.25±5.07	52.79±1.97	89.64±4.90	91.00±8.38	96.81±2.62	82.59±7.55	47.17±1.68	75.75±8.47	**97.41±1.86**	97.17±0.48
16	89.29±5.46	82.25±5.85	96.25±1.62	87.00±8.71	80.17±8.80	95.69±3.80	98.78±0.00	98.49±3.37	90.12±8.37	**100±0.00**
OA	80.30±0.29	79.65±0.45	94.12±1.25	95.00±0.77	92.48±2.10	92.68±0.71	73.48±0.70	88.86±0.89	96.86±0.36	**98.61±0.07**
AA	73.89±0.95	78.13±0.95	94.79±0.65	95.00±1.17	89.91±2.89	93.62±0.87	75.49±0.97	90.79±0.46	95.80±0.87	**99.01±0.07**
Kappa	77.41±0.36	76.61±0.54	93.25±1.44	95.00±0.89	91.62±2.47	91.66±0.81	69.45±0.80	87.29±1.00	96.42±0.41	**98.41±0.08**

In categories 2 and 11, the overlap is particularly evident in Figs. 12(a) and (d). It can be seen in Fig. 12(e) that our method can map samples evenly to a high-dimensional feature space to facilitate classification.

2) Pavia University Data Set: The classification results of each algorithm on the Pavia University data set are shown in Fig. 10. In Fig. 10(l), we can see the advantage of our method in an irregular shape from the correct classification of category 6 and category 7. From Fig. 10(g) and (h), the classification effect of category 7 is poor because the size of the convolution kernel based on CNN is fixed, and hence, this method cannot adapt to the situation of changeable shape.

The specific classification results for each algorithm are presented in Table VIII. Our method achieves the best classification results on OA, AA, and Kappa, and OA reaches 99.44%. Compared with SVM-3DG, our method is 2.01% higher in AA. SVM-3DG transforms spectral information and ignores the relationship between spectra. However, our method focuses on the spectrum information, which shows the superiority of our KSAM-SID.

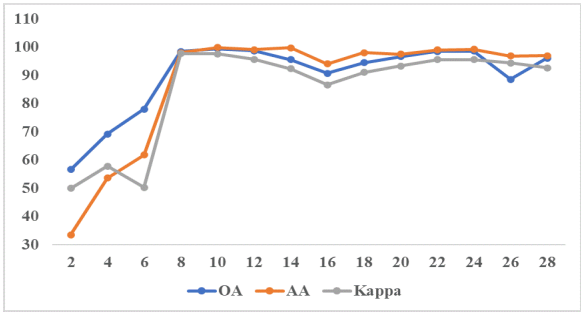


Fig. 8. Comparison of the results of different layers.

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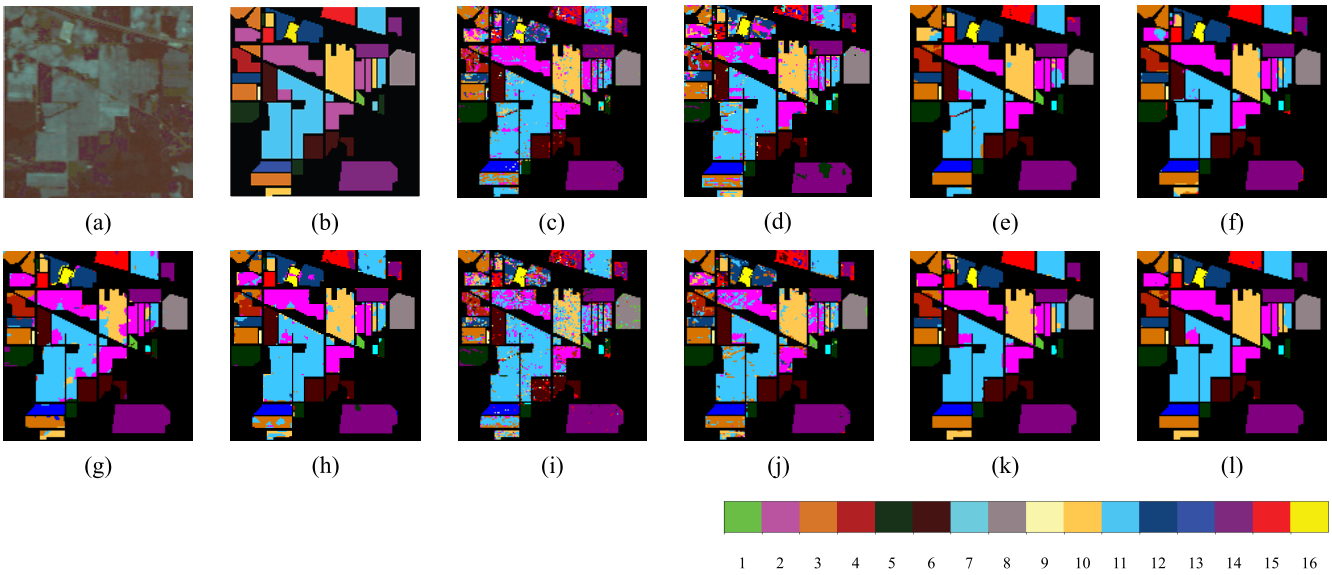


Fig. 9. Comparison of experimental classification results on Indian Pines data set. (a) Source image. (b) Ground truth. (c) SVM. (d) SAE-LR. (e) EPF. (f) SVM-3DG. (g) 2-D-CNN. (h) 3-D-CNN. (i) GCN. (j) FuNet-C. (k) MDGCN. (l) DAGCN.

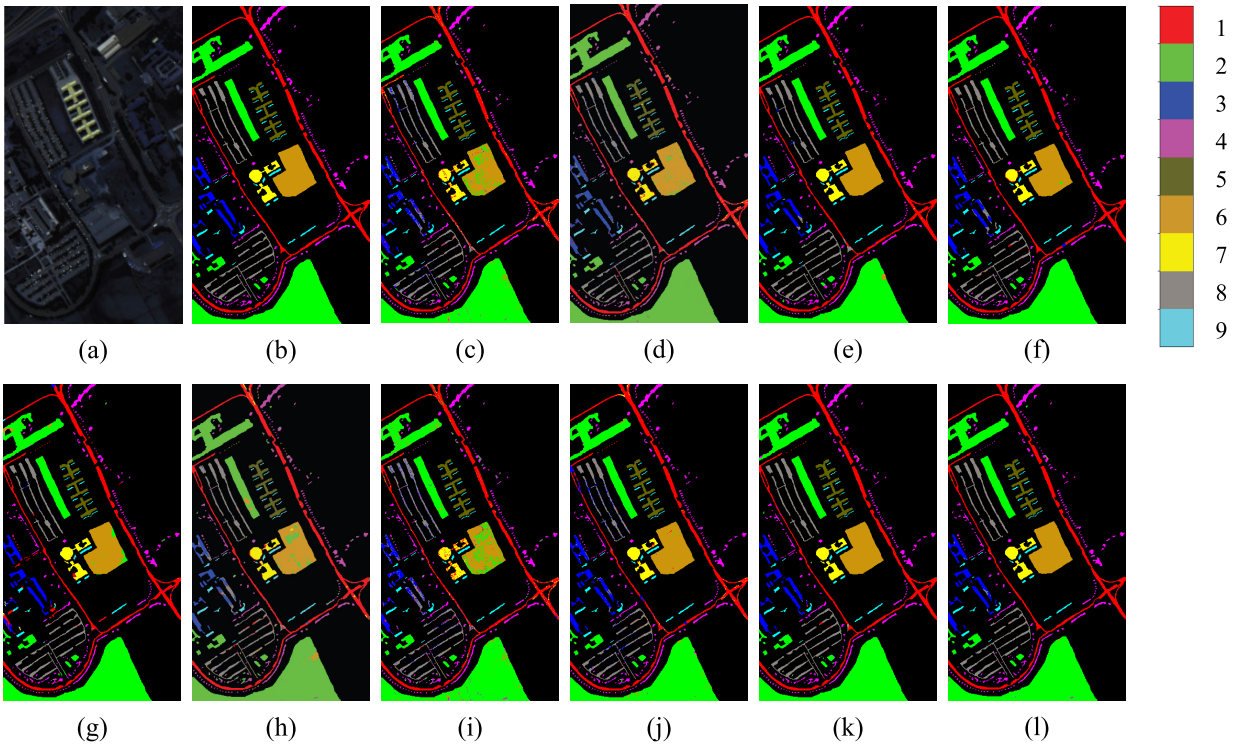


Fig. 10. Comparison of experimental classification results on Pavia University. (a) Source image. (b) Ground truth. (c) SVM. (d) SAE-LR. (e) EPF. (f) SVM-3DG. (g) 2-D-CNN. (h) 3-D-CNN. (i) GCN. (j) FuNet-C. (k) MDGCN. (l) DAGCN.

The second line of Fig. 12 is the T-SNE feature visualization of the Pavia University data set. In Fig. 12(f)–(h), category 2 and category 6 overlap seriously. Categories 1, 2, and 3 in Fig. 12(i) are mixed together. In contrast, the overlap of T-SNE features of our method is less.

3) Salinas Data Set: The classification results of each algorithm on the Salinas data set are shown in Fig. 11. It is obvious that, compared with other methods, our method has the best effect in categories 8 and 15. We find that, in categories 8 and 11, all the methods are not very effective, indicating that the two categories are difficult to distinguish. However, our method reduces errors by the KSAM-SID similarity measurement.

The specific classification results for each algorithm are shown in Table IX. Our method achieves the best results in OA, AA, and Kappa evaluation indexes. Among all the comparison methods, SVM-3DG has the best classification effect, followed by MDGCN, while the commonly used 2-D-CNN and 3-D-CNN are not very good. It can be seen that simple feature extraction using convolution cannot achieve

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TABLE VIII
COMPARISON OF EXPERIMENTAL CLASSIFICATION RESULTS ON PAVIA UNIVERSITY

Class No.	SVM	SAE-LR	EPF	SVM-3DG	2D-CNN	3D-CNN	GCN	FuNet-C	MDGCN	DAGCN
1	93.20±0.93	93.59±0.24	99.65±0.26	**99.73±0.20**	98.27±0.78	96.77±1.52	92.49±0.12	96.77±1.85	99.18±0.22	99.70±0.02
2	98.29±0.05	98.02±0.25	99.82±0.05	**99.82±0.09**	98.55±0.64	97.08±3.37	95.66±0.07	99.57±0.40	99.75±0.12	99.74±0.01
3	78.18±2.07	80.78±0.98	94.34±3.70	87.88±4.30	84.37±10.27	90.52±4.49	68.43±0.24	92.03±2.36	97.66±1.06	**98.34±0.09**
4	93.88±1.04	96.28±0.31	**99.48±0.48**	98.82±0.39	90.79±2.09	94.54±1.68	92.44±0.13	98.47±0.65	95.92±0.83	99.09±0.05
5	99.23±0.46	99.90±0.03	99.62±0.33	98.88±0.56	99.37±0.78	99.42±0.93	99.76±0.00	100±0.00	98.32±1.04	**100±0.00**
6	85.98±0.40	89.19±0.89	93.97±2.98	99.42±0.56	94.92±3.19	93.36±8.01	75.21±0.21	98.75±1.35	**100±0.00**	99.60±0.02
7	83.47±1.88	81.52±0.62	95.12±5.18	94.66±2.74	89.54±7.39	91.23±3.74	76.37±0.28	90.17±4.25	98.90±0.86	**99.13±0.06**
8	90.38±1.12	90.82±0.70	94.05±1.91	97.20±1.14	**97.96±1.05**	92.24±5.74	83.77±0.29	92.08±1.83	97.52±0.99	97.91±0.06
9	99.84±0.17	99.95±0.10	100±0.00	99.02±0.88	80.89±7.57	99.49±0.45	97.95±0.10	99.98±0.05	92.36±1.97	**100±0.00**
OA	93.67±0.20	94.22±0.29	98.12±0.72	98.67±0.20	96.13±0.52	95.62±1.27	89.75±0.10	97.68±0.48	98.89±0.08	**99.44±0.02**
AA	91.38±0.33	92.23±0.30	97.34±1.17	97.27±0.33	92.74±0.86	94.96±1.47	86.90±0.11	96.43±0.52	97.73±0.15	**99.28±0.02**
Kappa	91.56±0.27	92.40±0.38	97.47±0.97	98.24±0.26	94.86±0.69	94.21±1.66	86.33±0.14	96.92±0.63	98.52±0.11	**99.26±0.02**

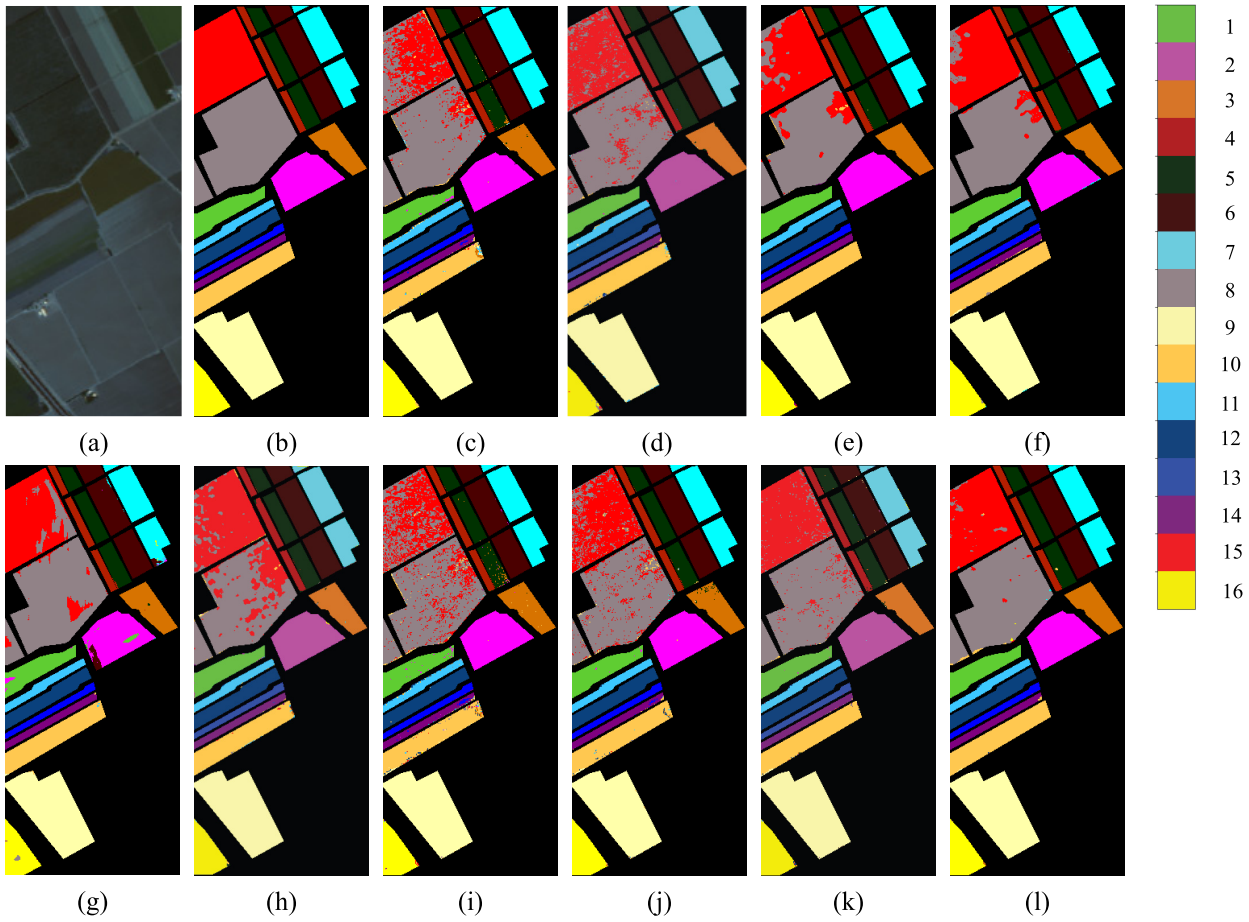


Fig. 11. Comparison of experimental classification results on the Salinas data set. (a) Source image. (b) Ground truth. (c) SVM. (d) SAE-LR. (e) EPF. (f) SVM-3DG. (g) 2-D-CNN. (h) 3-D-CNN. (i) GCN. (j) FuNet-C. (k) MDGCN. (l) DAGCN.

good results. Our similarity measurement method provides an effective guarantee for accurate classification, which can be seen from the OA rate of 99.04%.

The third line of Fig. 12, namely, Fig. 12(k)–(o), presents the T-SNE feature visualization of the Salinas data set. Looking at the T-SNE features, it is obvious that the overlap between categories 8 and 15 is the most serious. In Fig. 12(o), although our method still has little overlap in categories 8 and 15, it largely distinguishes the two categories.

4) Houston Data Set: The classification results on the Houston data set are shown in Fig. 13. By observing Fig. 13(b) and (c), we find that the classification result of our method is basically consistent with that of ground truth.

The specific classification results for each algorithm are shown in Table X. It can be seen that our method achieves the best results on OA, AA, and Kappa. Compared with the EPF with the best results in other methods, our OA is improved by 1.93%, AA by 1.73%, and Kappa by 2.09%, respectively.

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TABLE IX
COMPARISON OF EXPERIMENTAL CLASSIFICATION RESULTS ON SALINAS

Class No.	SVM	SAE-LR	EPF	SVM-3DG	2D-CNN	3D-CNN	GCN	FuNet-C	MDGCN	DAGCN
1	98.42±0.82	99.22±0.21	100±0.00	99.08±0.81	94.48±3.12	99.26±0.98	97.91±0.05	99.74±0.23	97.47±0.36	**100±0.00**
2	99.83±0.10	99.37±0.14	100±0.00	99.50±0.42	95.23±2.81	99.61±0.28	99.97±0.00	99.9±0.05	93.20±3.37	**100±0.00**
3	99.44±0.23	99.22±0.09	98.99±1.37	98.95±0.76	99.67±0.52	94.56±5.98	99.39±0.03	98.31±1.41	99.55±0.76	**100±0.00**
4	99.50±0.15	99.66±0.21	98.80±0.54	98.29±0.90	96.20±5.30	99.01±0.67	99.49±0.03	99.57±0.52	99.06±1.65	**99.87±0.04**
5	97.53±0.69	99.38±0.24	**99.97±0.02**	99.89±0.22	96.96±3.23	98.10±1.42	95.85±0.03	97.74±2.05	99.18±0.94	99.57±0.00
6	99.75±0.14	99.95±0.04	100±0.00	99.96±0.03	97.80±3.59	99.87±0.13	99.82±0.01	99.95±0.05	99.72±0.25	**100±0.00**
7	99.49±0.28	99.58±0.04	**100±0.00**	99.28±0.77	98.80±1.41	99.62±0.19	99.87±0.03	99.9±0.07	99.85±0.14	99.89±0.02
8	91.76±0.17	86.60±0.29	91.10±3.09	96.00±0.92	88.27±8.56	86.22±2.52	84.86±0.09	88.61±2.06	90.43±5.53	**98.34±0.03**
9	99.69±0.40	99.14±0.09	99.97±0.04	99.74±0.28	99.28±0.90	99.14±0.65	99.81±0.01	99.19±1.02	99.31±0.53	**100±0.00**
10	93.66±1.01	94.58±0.23	95.76±1.86	97.76±1.16	99.61±0.28	93.56±5.43	93.32±0.05	97.09±0.33	98.99±0.46	**99.63±0.01**
11	94.54±2.25	91.03±0.60	99.19±0.94	99.10±0.25	**99.74±0.26**	92.15±4.71	91.09±0.17	98.3±1.95	99.55±0.39	98.97±0.05
12	99.95±0.04	99.48±0.20	99.98±0.02	99.89±0.16	99.95±0.04	99.40±0.54	99.84±0.00	100±0.00	99.82±0.25	**100±0.00**
13	98.46±0.50	99.97±0.05	**100±0.00**	98.72±0.59	99.70±0.32	99.17±0.81	98.65±0.05	99.87±0.12	99.79±0.35	99.66±0.00
14	93.16±0.83	97.96±0.42	99.28±0.55	96.55±3.16	**99.78±0.22**	96.78±2.04	90.22±0.25	96.00±1.27	99.76±0.32	98.54±0.12
15	58.59±1.41	76.68±0.81	87.89±1.77	86.74±2.86	94.04±6.83	75.12±13.81	57.46±0.16	82.14±3.68	**96.95±3.79**	96.33±0.05
16	98.65±0.22	95.37±0.95	**99.95±0.07**	98.77±1.10	98.65±1.56	97.84±2.55	98.59±0.05	99.20±0.34	98.97±1.26	99.39±0.03
OA	91.71±0.23	93.15±0.10	96.08±0.83	96.86±0.29	95.56±1.98	92.57±2.12	89.92±0.05	94.61±0.39	96.73±1.09	**99.04±0.02**
AA	95.15±0.23	96.07±0.07	98.18±0.38	98.01±0.30	97.38±0.86	95.59±1.75	94.14±0.05	97.22±0.36	98.22±0.21	**99.39±0.01**
Kappa	90.74±0.26	92.36±0.12	95.62±0.94	96.50±0.33	95.06±2.20	91.72±2.38	88.75±0.06	94.00±0.44	96.36±1.21	**98.93±0.02**

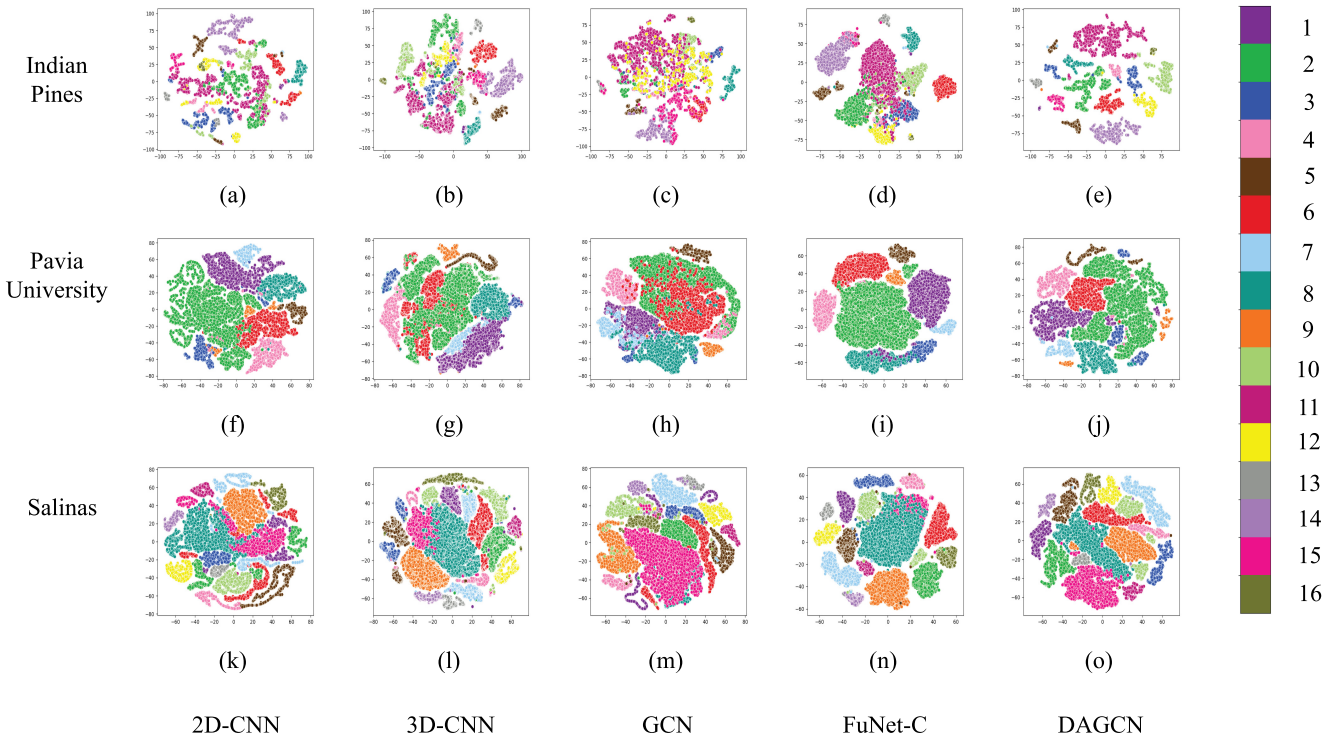


Fig. 12. T-SNE features visualization of hyperspectral data set. T-SNE features visualization of hyperspectral data set. T-SNE denote T-SNE features of (a)–(e) Indian Pines dataset, (f)–(j) Pavia University dataset, and (k)–(o) Salinas dataset. From left to right of each line are obtained based on the methods 2D-CNN, 3D-CNN, GCN, Funet-C, and DAGCN.

In summary, the proposed DAGCN achieves the best performance for the OA, AA, and Kappa coefficients in the three data sets, which proves the effectiveness and superiority of the proposed method. Experiments show that, when our training samples are small, even if there are only ten samples, the classification results are good. In the result figures for HIC, we see that DAGCN can solve the problem that irregular shapes are difficult to distinguish and that the interference between different categories is large. In addition, the classification results are more or less accurate in the narrow region and at the category boundary. This shows that our GCN structure based on spectral similarity can adapt to the non-Euclidean structure of HSIs. From the perspective of the T-SNE feature, our method can

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TABLE X
COMPARISON OF EXPERIMENTAL CLASSIFICATION RESULTS ON HOUSTON

Class No.	SVM	SAE-LR	EPF	SVM-3DG	2D-CNN	3D-CNN	GCN	FuNet-C	MDGCN	DAGCN
1	96.43±0.71	96.46±1.14	98.28±0.93	98.66±0.15	88.09±7.86	97.74±0.73	95.69±0.22	98.2±0.74	98.67±0.09	**99.56±0.80**
2	98.21±0.77	98.04±0.21	98.29±0.62	96.59±0.66	83.71±10.86	99.49±0.12	95.85±0.15	**99.66±0.17**	99.36±0.04	99.30±0.44
3	98.49±0.23	95.92±1.20	99.76±0.20	98.85±0.98	99.02±1.10	97.20±1.47	98.74±0.14	99.92±0.17	99.72±0.00	**99.97±0.06**
4	93.09±0.30	97.42±1.80	**99.47±0.70**	93.72±0.67	80.77±6.01	98.56±0.67	93.82±0.08	98.58±1.09	94.37±0.32	98.71±2.36
5	97.21±0.99	98.32±0.06	98.11±1.59	99.26±0.34	98.25±1.96	96.92±3.19	96.14±0.14	99.58±0.38	99.69±0.14	**99.84±0.11**
6	90.59±4.99	96.56±0.23	**100±0.00**	90.49±5.04	88.77±4.64	91.91±4.71	95.07±0.33	94.63±1.21	99.08±0.15	98.82±1.08
7	84.32±2.22	88.49±0.69	94.38±1.55	96.25±3.04	93.32±3.48	89.21±3.92	82.11±0.36	94.65±0.50	**98.81±0.08**	97.57±1.07
8	74.44±5.03	84.52±2.72	98.54±1.16	86.58±2.67	64.06±9.37	89.01±3.65	87.87±0.41	91.68±2.26	73.95±0.47	**97.70±1.31**
9	76.20±2.41	77.48±2.75	94.45±2.26	93.72±1.10	92.31±1.43	89.32±4.69	85.60±0.28	90.08±1.04	86.95±1.49	**97.45±1.63**
10	79.84±3.42	89.24±3.46	96.35±1.35	93.68±3.15	98.59±1.66	94.01±1.94	89.55±0.32	94.42±0.62	97.01±1.78	**99.80±0.32**
11	70.51±5.16	80.25±3.59	97.25±1.90	82.47±1.63	98.22±1.28	95.20±1.5	84.13±0.23	94.64±0.74	95.31±1.40	**99.25±0.57**
12	74.67±1.52	79.01±2.45	93.91±1.78	92.32±1.39	93.46±2.42	88.52±2.57	85.17±0.40	93.98±1.00	97.27±0.59	**98.77±0.60**
13	22.78±6.13	46.48±1.12	90.16±4.72	92.04±2.03	79.53±6.97	81.66±7.87	70.86±0.46	86.87±3.21	90.97±1.13	**96.61±1.97**
14	97.66±0.71	97.83±0.62	98.75±2.17	99.13±0.41	99.35±0.51	93.29±5.30	98.91±0.00	99.50±0.43	99.26±0.36	**100±0.00**
15	97.08±0.69	97.49±0.40	99.71±0.57	98.75±1.07	98.38±1.75	98.56±0.91	97.04±0.12	99.45±0.23	100±0.00	**100±0.00**
OA	83.65±0.29	88.20±0.64	96.92±0.38	93.79±0.64	90.17±2.37	93.59±0.85	89.87±0.18	95.62±0.17	94.81±0.11	**98.85±0.22**
AA	83.43±0.55	88.23±0.66	97.16±0.53	94.17±0.69	90.42±2.21	93.37±1.03	90.44±0.17	95.72±0.20	95.36±0.12	**98.89±0.21**
Kappa	82.30±0.32	87.23±0.69	96.67±0.41	93.29±0.69	89.37±2.56	93.07±0.92	89.05±0.19	95.26±0.19	94.38±0.12	**98.76±0.24**

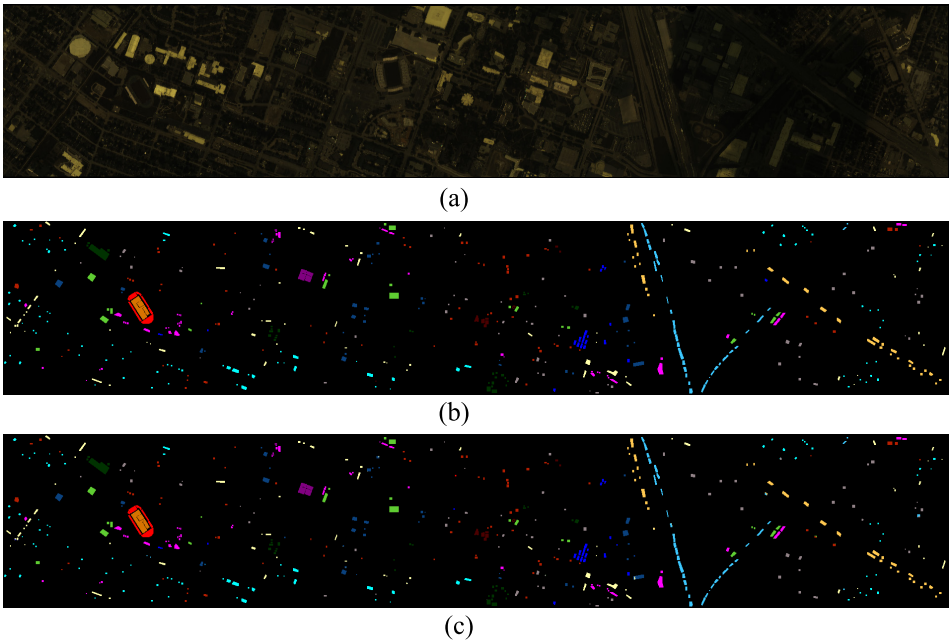


Fig. 13. Experimental classification result on the Houston data set. (a) Source image. (b) Ground truth. (c) DAGCN.

uniformly map hyperspectral data to a high-dimensional feature space, there is less overlap between samples, and the same category has certain aggregation. This implies that our DeepGCN structure can better map hyperspectral data into high-dimensional space, which is more suitable for nonlinear classification.

V. CONCLUSION

In this article, we developed an HIC method by designing the DAGCN. Specifically, a new KSAM-SID similarity measurement method is used to construct an adjacency matrix that considers spectral and spatial location information. Then, according to the attention learning mechanism, the samples are given different characteristics of aggregate influence. From the perspective of network design, to solve the problem that increasing the number of convolution layers results in the indivisibility of features and gradient disappearance, a deep graph CNN is designed by utilizing dense connections and dilated convolution design techniques. Furthermore, the graph adjacency matrix is updated dynamically, while the network is trained. Experiments show that the proposed method can effectively extract the hyperspectral data features and improve classification accuracy.

Our findings show that arbitrary shape convolution is more suitable for HIC at the pixel level. Due to the new similarity measurement method, sparse matrix storage can be used to save space although it uses all spectral information. In addition, we realize that the computational complexity is still high,

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especially when the hyperspectral data set is large. Hence, end-to-end learning is a great challenge. Seeking solutions in future work, we will investigate design methods for automatic end-to-end representation learning and implement large-scale data learning.

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